

**B.Tech. Degree II Semester Regular and I Semester Supplementary
Examination May 2016**

**IT/CS/EC/CE/EE/ME/SE 1103 A / 1203 B ENGINEERING MECHANICS
(2015 Scheme)**

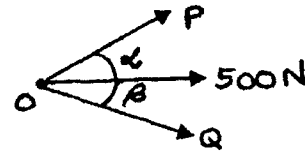
Time : 3 Hours

Maximum Marks : 60

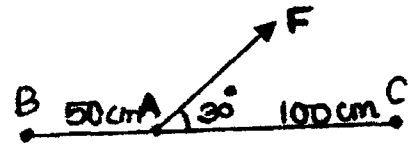
**PART A
(Answer ALL questions)**

(10 × 2 = 20)

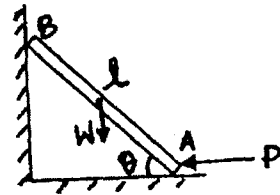
- I. (a) If the resultant of the two forces P and Q is 500 N and if the angles $\alpha = 30^\circ$ and $\beta = 45^\circ$, what are the magnitude of P and Q?



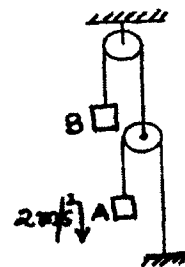
- (b) Explain briefly the principle of superposition and transmissibility.
(c) A force $F = 250$ N acting at A is to be resolved into two parallel components acting at B and C. Determine the components.



- (d) State and prove parallel axis theorem of moment of inertia.
(e) A slender prismatic bar AB of length 'l' and weight W stands in a vertical plane and it is supported by smooth surfaces at A and B as shown. Using the principle of virtual work, find the horizontal force P applied at A to keep the bar in equilibrium.



- (f) A car travels with a velocity $v = 20(1 - e^{-t})$ m/s where t is the elapsed time in seconds. Determine the distance travelled in 3s.
(g) Determine the acceleration of block A, if the acceleration of block B is 2 m/s^2 .

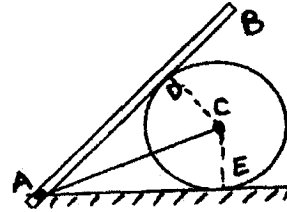


- (h) State and prove the impulse-momentum principle in rectilinear translation.
- (i) Derive the expression for the period of oscillation of a compound pendulum.
- (j) A cord is wrapped around a wheel of radius of 0.1m initially at rest. If a force is applied to the cord and gives it an acceleration $a = 4t \text{ m/s}^2$ (where 't' is in seconds), determine the angular velocity of the drum as a function of time.

PART B

(4 × 10 = 40)

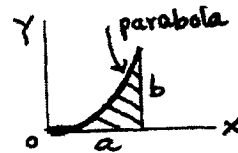
- II. (a) A roller of radius $r = 50 \text{ cm}$ rests on a horizontal plane and is held in equilibrium position by a string AC of length 100 cm. A heavy bar AB of weight $W = 500 \text{ N}$ and length $l = 150 \text{ cm}$ is hinged at A and rests on the roller as shown. Neglecting friction in all contact surfaces, find the tension in the string and the reaction on the roller by the ground.



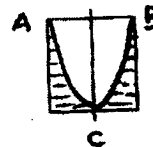
- (b) A uniform ladder of length $L = 3 \text{ m}$ and weight W leans against a vertical wall at an angle of 45° . If the coefficient of friction between the wall and ladder is 0.3 and that between the ladder and ground is 0.5, how high a man of weight $W/2$ can go up the ladder before it slips?

OR

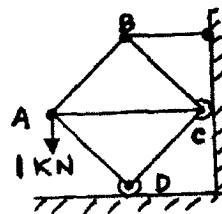
- III. (a) Determine the y co-ordinate of the centroid of the parabolic spandrel shown in figure.



- (b) A right circular cylindrical tank containing water spins about its vertical axis at such speed that the free water surface is a paraboloid of ACB. What will be the height of water when it comes to rest?

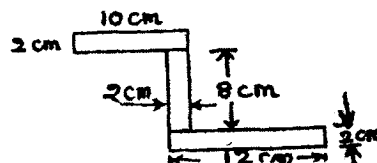


- IV. Determine the axial force in the bars of the truss shown in figure. ABCD is a square with AC horizontal.



OR

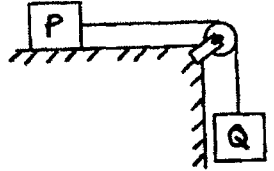
- V. Determine the moment of inertia of the z-section shown in figure with respect to the centroidal axes.



- VI. (a) The driver of an automobile, travelling along a straight level highway, suddenly applies brakes so that the car skids for 2s covering a distance of 10 m before coming to rest. Assuming that during this time the car moves with constant deceleration, find the coefficient of friction between the tyres and the road.
- (b) A wooden block of weight 50 N rests up on a rough horizontal plane, the coefficient of friction between the two being 0.4. If a bullet weighting 0.2 N is fired horizontally in to the block with a muzzle velocity $v = 600\text{ m/s}$, how far will the block be displaced from its position, assuming the bullet remains inside the block?

OR

- VII. (a) Two blocks of weights $P = 50\text{ N}$ and $Q = 25\text{ N}$ are connected by a flexible but inextensible cord and supported as shown. If the coefficient of friction between the block P and the horizontal surface is 0.3, and all other friction is neglected, find the acceleration of the falling weight Q and the tension in the string.
- (b) Two springs shown in figure each have a spring constant k and the attached ball has a weight W . If the ball is initially displaced to the right and released, find the period of oscillation of the ball. Neglect friction.



- VIII. (a) A car travels around a horizontal circular track that has a radius of 90 m. If the car increases its speed at a constant rate of 2 m/s^2 , determine the time needed for it to reach a resultant acceleration of 3 m/s^2 .
- (b) Two adjacent guns fire bullets at 300 m/s simultaneously at angles of elevation α_1 and α_2 for the same target at a range of 200 m. Calculate the time difference between the two hits.

OR

- IX. (a) A homogeneous circular disc of radius r and weight W hangs in a vertical plane from a pin O at its circumference. Find the period for small oscillations of the disc.
- (b) A solid right circular drum of radius $r = 30\text{ cm}$ and weight $W = 150\text{ N}$ is free to rotate about its geometric axis. A flexible cord carrying a weight of $Q = 50\text{ N}$ is wound around the circumference of the drum. If the weight Q is released from rest, find the time t required for it to fall through a height $h = 3\text{ m}$. With what velocity will it strike the floor?

